Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

INFLUENCER RANKING AND ENGAGEMENT ANALYSIS

The domain of the Project:

Full Stack Development

Team Mentors (and their designation):

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Team Members:

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Period of the project

March 2026 to April 2026

ABSTRACT

The **Influencer Ranking and Engagement Analysis Platform** is a full-stack web application designed to help marketers and businesses identify the most effective social media influencers based on real audience engagement rather than superficial metrics like follower count. In today's digital marketing landscape, influencer marketing plays a crucial role in brand promotion, but selecting the right influencers remains a major challenge due to the lack of reliable and data-driven evaluation methods.

This project aims to bridge this gap by providing a scalable, efficient, and user-friendly platform that analyzes influencer performance using meaningful engagement metrics. The system is developed using the MERN stack (MongoDB, Express.js, React.js, and Node.js), ensuring high performance, flexibility, and scalability. The frontend is built with React.js to deliver a dynamic and responsive user interface, while the backend uses Node.js and Express.js to handle API operations and data processing. MongoDB is used as the database to store influencer data and engagement metrics efficiently.

The platform collects real-time Instagram data using the Apify API based on user-provided hashtags. It processes the collected data to extract important features such as likes, comments, follower count, and posting frequency. A custom Engagement Score algorithm is implemented to evaluate influencers by normalizing engagement metrics, ensuring a fair and accurate ranking system.

The system provides an interactive dashboard where users can view ranked influencers, apply filters, and export results for further analysis. It supports data-driven decision-making, reduces manual effort, and improves marketing campaign effectiveness.

CHAPTER-1

INTRODUCTION

1.1 Introduction

In the modern digital era, social media has become a powerful platform for communication, marketing, and brand promotion. Among various strategies, influencer marketing has emerged as one of the most effective ways for businesses to reach targeted audiences. Influencers, who have a strong online presence and audience trust, play a key role in shaping consumer decisions. However, identifying the right influencers for marketing campaigns remains a challenging task.

Most traditional approaches rely heavily on basic metrics such as follower count, which does not accurately represent an influencer's true impact. High follower numbers do not always guarantee meaningful audience engagement, making it difficult for marketers to select suitable influencers. Therefore, there is a need for a more reliable, data-driven system that evaluates influencers based on actual engagement rather than popularity alone.

This project introduces an **Influencer Ranking and Engagement Analysis Platform**, which automates the process of influencer evaluation using real-time data and analytical techniques. The system focuses on measuring genuine engagement through metrics such as likes, comments, and posting activity, providing a more accurate representation of influencer performance.

Overall, this project offers a modern, automated, and intelligent solution for influencer selection, transforming raw social media data into actionable insights. It also demonstrates practical implementation of full-stack development and data analysis concepts, making it highly valuable for both academic and real-world applications.

1.2 Project Background

With the rapid growth of platforms like Instagram, influencer marketing has become a core component of digital advertising strategies. Businesses collaborate with influencers to promote products, increase brand awareness, and drive customer engagement. However, the process of selecting influencers is often manual, time-consuming, and prone to errors.

Existing methods typically involve searching hashtags, analyzing profiles individually, and maintaining data in spreadsheets. This approach lacks efficiency and does not provide a standardized way to compare influencers. Additionally, many third-party tools focus mainly on surface-level analytics and fail to capture deeper engagement insights.

To address these challenges, modern web technologies such as the MERN stack (MongoDB, Express.js, React.js, Node.js) can be used to build scalable and efficient applications. By integrating real-time data collection tools like the Apify API, it becomes possible to automate data gathering and analysis. This project leverages these technologies to create a comprehensive platform that simplifies influencer evaluation and improves decision-making.

1.3 Problem Statement

Despite the growing importance of influencer marketing, several challenges exist in the current system:

- Influencer selection is largely manual and time-consuming

- Heavy reliance on follower count leads to inaccurate evaluation
- Lack of standardized metrics for comparing influencers
- Limited access to real-time engagement data
- Poor decision-making resulting in ineffective marketing campaigns

There is a need for a system that can:

- Automatically collect influencer data
- Analyze engagement using meaningful metrics
- Provide accurate ranking based on real performance
- Support efficient and data-driven decision-making

1.4 Objectives

The main objective of this project is to design and develop a web-based platform that ranks influencers based on engagement analysis rather than superficial metrics.

The specific objectives include:

- To develop a full-stack web application using the MERN stack
- To collect real-time social media data using APIs
- To analyze engagement metrics such as likes, comments, and post frequency
- To implement a custom Engagement Score algorithm for ranking influencers
- To provide an interactive dashboard for visualization and filtering
- To enable export of analyzed data for reporting purposes
- To ensure scalability, performance, and user-friendly design

1.5 Benefits of the Proposed Platform

The proposed system offers several advantages for different users:

- **Marketers:** Helps in selecting the most effective influencers based on real engagement
- **Businesses:** Improves marketing campaign performance and ROI
- **Researchers/Students:** Provides practical insights into data analysis and social media analytics
- **Developers:** Demonstrates real-world implementation of full-stack development

Overall, the platform reduces manual effort, enhances accuracy in influencer selection, and supports data-driven marketing strategies. It also serves as a valuable academic project showcasing modern web technologies and analytical approaches.

CHAPTER-2

SCOPE OF THE PROJECT

2.1 Scope

The scope of the **Influencer Ranking and Engagement Analysis Platform** includes the complete process of collecting, analyzing, and ranking social media influencers based on engagement metrics. The system is designed to support users such as marketers, researchers, and businesses in identifying the most suitable influencers for marketing campaigns.

The platform allows users to input hashtags related to specific domains or industries. Based on these inputs, the system collects real-time Instagram data using the Apify API. The collected data includes influencer profiles, posts, likes, comments, and other engagement-related information.

The system processes this data to extract meaningful features such as average likes, comments, engagement rate, and posting frequency. These features are then used to calculate a custom Engagement Score, which helps rank influencers based on actual audience interaction.

The platform also provides a user-friendly dashboard where users can:

- View ranked influencers
- Apply filters based on engagement criteria
- Analyze performance using visual insights
- Export results for reporting and further analysis

Overall, the system covers the entire workflow from data collection to

actionable insights, making influencer selection more efficient and reliable.

2.2 Functional Scope (Modules)

2.2.1 User Authentication Module

- Allows users to register and log in securely
- Manages user sessions and authentication
- Ensures secure access to dashboard and features
- Protects user data using proper validation techniques

2.2.2 Hashtag Input & Data Collection Module

- Allows users to input hashtags based on niche or category
- Collects real-time Instagram data using the Apify API
- Retrieves influencer profiles, posts, likes, and comments
- Acts as the primary source of data for analysis

2.2.3 Data Processing & Feature Extraction Module

- Cleans and filters raw collected data
- Removes duplicate or irrelevant data
- Extracts important features such as:
  - Average likes
  - Average comments
  - Follower count
  - Engagement rate
  - Posting frequency
- Converts raw data into structured analytical data

2.2.4 Engagement Score Calculation Module

- Calculates a custom Engagement Score for each influencer
- Normalizes likes and comments based on follower count

- Includes posting frequency using weighted formulas

- Ensures fair evaluation based on real engagement

2.2.5 Influencer Ranking & Result Module

- Ranks influencers based on Engagement Score
- Displays results in descending order of performance
- Provides filtering options for better analysis
- Enables exporting results for reports

2.3 Uniqueness and Project Highlights

This project stands out due to its focus on **data-driven influencer evaluation** rather than traditional methods:

- **Engagement-Based Ranking:** Prioritizes real audience interaction over follower count
- **Automation:** Eliminates manual effort in data collection and analysis
- **Real-Time Data:** Uses APIs to fetch up-to-date influencer data
- **Interactive Dashboard:** Provides visualization and filtering features
- **Scalable Architecture:** Built using MERN stack for flexibility and future expansion
- **Practical Relevance:** Useful for both academic learning and real-world marketing.

CHAPTER-3

EXISTING SYSTEM VS PROPOSED SYSTEM

3.1 Existing System

In the current digital marketing environment, influencer selection is mostly carried out using manual and unstructured methods. Marketers typically rely on social media platforms like Instagram to search for influencers using hashtags and then evaluate profiles individually.

Most existing approaches depend heavily on surface-level metrics such as follower count, which does not accurately reflect an influencer's true engagement or impact. Additionally, many third-party tools provide only basic analytics and often require paid subscriptions to access advanced features.

The overall workflow involves manual data collection, spreadsheet-based analysis, and subjective decision-making. This process is time-consuming, inefficient, and prone to human error. Furthermore, there is no standardized method to compare influencers based on meaningful engagement metrics, making it difficult to identify the most suitable candidates for marketing campaigns.

3.2 Proposed System

The proposed system introduces an automated and data-driven platform for influencer ranking and engagement analysis. It eliminates manual effort by collecting real-time Instagram data using the Apify API based on user-defined hashtags.

The system processes the collected data to extract key engagement metrics such as likes, comments, follower count, and posting frequency. A custom Engagement Score algorithm is implemented to evaluate influencers based on normalized engagement rather than follower count.

The platform provides an interactive dashboard where users can:

- View ranked influencers
- Apply filters based on engagement criteria
- Analyze performance through visual insights
- Export results for reporting

This approach ensures accurate, efficient, and objective influencer selection, enabling better marketing decisions and improved campaign performance.

3.3 Comparison Table

Existing System	Proposed System
Manual influencer search using hashtags	Automated data collection using Apify API
Relies on follower count for evaluation	Uses engagement-based ranking system
Time-consuming and inefficient process	Fast and automated workflow
No standardized evaluation method	Custom Engagement Score for accurate comparison
Data stored in spreadsheets	Centralized database using MongoDB
Limited analytics and insights	Advanced filtering and visualization dashboard
High risk of human error	Reduced errors through automation

Existing System	Proposed System
Requires manual decision-making	Supports data-driven decision-making

3.4 Feature Summary (What I Implement)

The key features implemented in this project include:

- Real-time data collection using hashtags
- Engagement-based influencer ranking system
- Custom Engagement Score algorithm
- Interactive dashboard for visualization
- Filtering options based on engagement metrics
- Export functionality for reports
- Scalable MERN stack architecture

CHAPTER-4

ARCHITECTURAL DIAGRAM

4.1 Architecture Overview

The **Influencer Ranking and Engagement Analysis Platform** is designed using a layered client-server architecture. This architecture divides the system into multiple layers to ensure better organization, scalability, and maintainability. The main layers include the User Layer, Frontend Layer, Backend Layer, and Database Layer.

This structured approach allows each component to function independently while maintaining smooth communication between layers. It improves system performance and makes it easier to update or extend individual components without affecting the entire system.

The application follows a request-response model, where user actions are processed through the frontend, handled by backend APIs, and stored or retrieved from the database. This architecture ensures efficient data processing, security, and scalability.

4.2 Architecture Description

4.2.1 User Layer

The user layer represents the end-users who interact with the system through web browsers. Users can access the platform to search for influencers, input hashtags, and view ranked results.

Authenticated users can use advanced features such as filtering results, exporting data, and accessing dashboards. This layer focuses on providing a smooth and user-friendly experience.

4.2.2 Frontend Layer (React.js)

The frontend is developed using React.js, which provides a dynamic and responsive user interface. It is responsible for handling user interactions and displaying data in a structured format.

Key functionalities of the frontend include:

- Input forms for hashtags
- Dashboard for displaying ranked influencers
- Filters for refining search results
- Visualization of engagement metrics

React uses a component-based architecture, allowing reusable UI components and efficient state management. The frontend communicates with the backend through HTTP requests.

4.2.3 Backend Layer (Node.js + Express.js)

The backend is implemented using Node.js and Express.js, which handle application logic and API communication.

Main responsibilities include:

- Handling API requests from the frontend
- Integrating with the Apify API to fetch Instagram data
- Processing and analyzing engagement metrics
- Implementing the Engagement Score algorithm
- Managing authentication and security

The backend acts as a bridge between the frontend and the database, ensuring proper data flow and processing.

4.2.4 Database Layer (MongoDB)

The database layer uses MongoDB to store all application data in a structured format.

The database stores:

- User details and authentication data
- Influencer profiles and engagement metrics
- Processed data for ranking and analysis

MongoDB provides flexibility, scalability, and efficient data retrieval, making it suitable for handling large volumes of social media data.

4.3 System Flow

The overall system flow can be described as:

User → React Frontend → Express API → Data Processing → MongoDB → Response to Frontend

When a user enters a hashtag:

1. The request is sent from the frontend to the backend
2. The backend fetches data using the Apify API
3. Data is processed and engagement scores are calculated
4. Results are stored in the database
5. Ranked influencers are sent back to the frontend
6. The frontend displays the results to the user

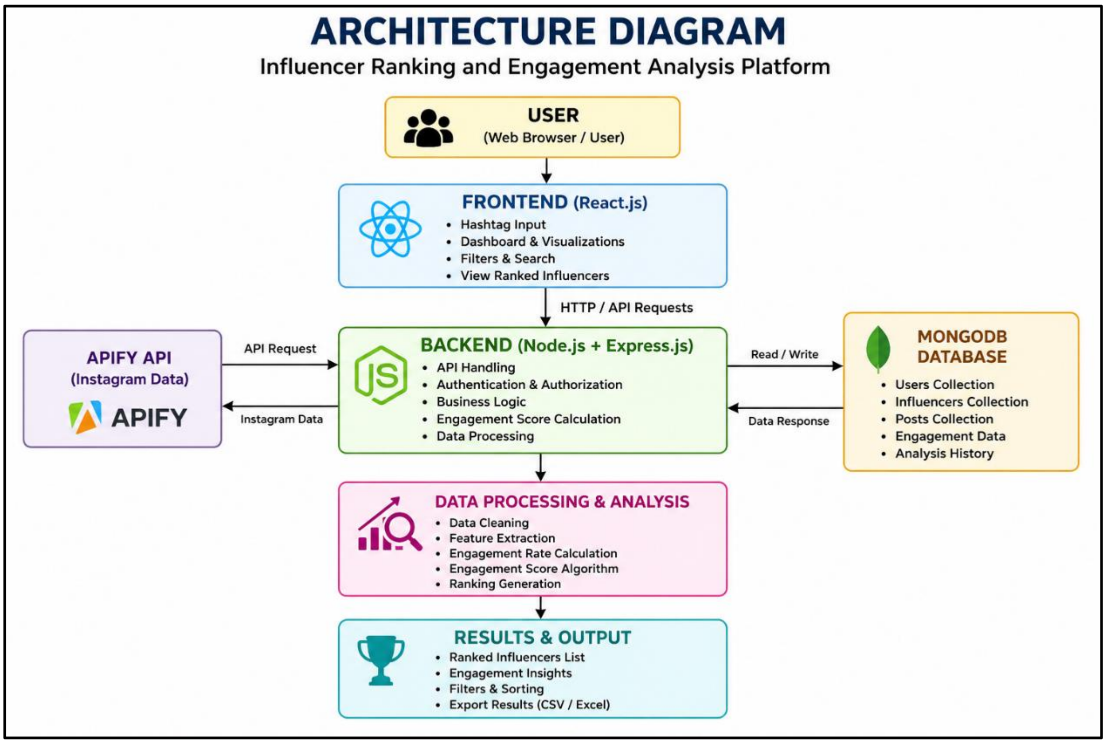
4.4 Advantages of the Architecture

The chosen architecture offers several benefits:

- Clear separation of frontend, backend, and database
- Improved scalability and flexibility
- Efficient handling of large datasets
- Secure data processing and storage
- Easy integration of new features and modules
- Better performance through optimized data flow

4.5 Architecture Diagram

The architecture diagram represents the interaction between all system components, including the user interface, backend server, API integration, and database.



System Architecture

CHAPTER-5

ER DIAGRAM

5.1 ERD Explanation

The **Entity Relationship Diagram (ERD)** represents the logical structure of the database used in the Influencer Ranking and Engagement Analysis Platform. It shows how different entities are related and how data flows within the system.

The ERD helps in designing a well-structured database by defining entities, their attributes, and relationships. It ensures data consistency, avoids redundancy, and supports efficient data retrieval.

The main entities in this system include **User, Influencer, Post, and Engagement**. These entities work together to support functionalities such as data collection, processing, ranking, and user interaction.

5.1.1 User Entity

The User entity stores information related to registered users of the platform.

Attributes:

- User ID (Primary Key)
- Name
- Email
- Password (hashed)
- Created Date

Each user can search for influencers, apply filters, and export results. Authentication is handled securely using stored credentials.

5.1.2 Influencer Entity

The Influencer entity contains details about social media influencers collected from Instagram.

Attributes:

- Influencer ID (Primary Key)
- Username
- Follower Count
- Following Count
- Total Posts
- Category/Niche

Each influencer is analyzed and ranked based on engagement metrics.

5.1.3 Post Entity

The Post entity represents the content posted by influencers.

Attributes:

- Post ID (Primary Key)
- Influencer ID (Foreign Key)
- Caption
- Number of Likes
- Number of Comments
- Post Date

Each influencer can have multiple posts, which are used to calculate engagement metrics.

5.1.4 Engagement Entity

The Engagement entity stores calculated engagement-related data.

Attributes:

- Engagement ID (Primary Key)

- Influencer ID (Foreign Key)
- Average Likes
- Average Comments
- Engagement Rate
- Engagement Score

This entity is used to rank influencers based on performance.

5.2 Relationships

The relationships between entities are defined as follows:

- **User → Influencer (One-to-Many):** A user can analyze multiple influencers.
- **Influencer → Post (One-to-Many):** One influencer can have multiple posts.
- **Influencer → Engagement (One-to-One):** Each influencer has one engagement record.
- **Post → Engagement (Many-to-One):** Multiple posts contribute to one engagement calculation.

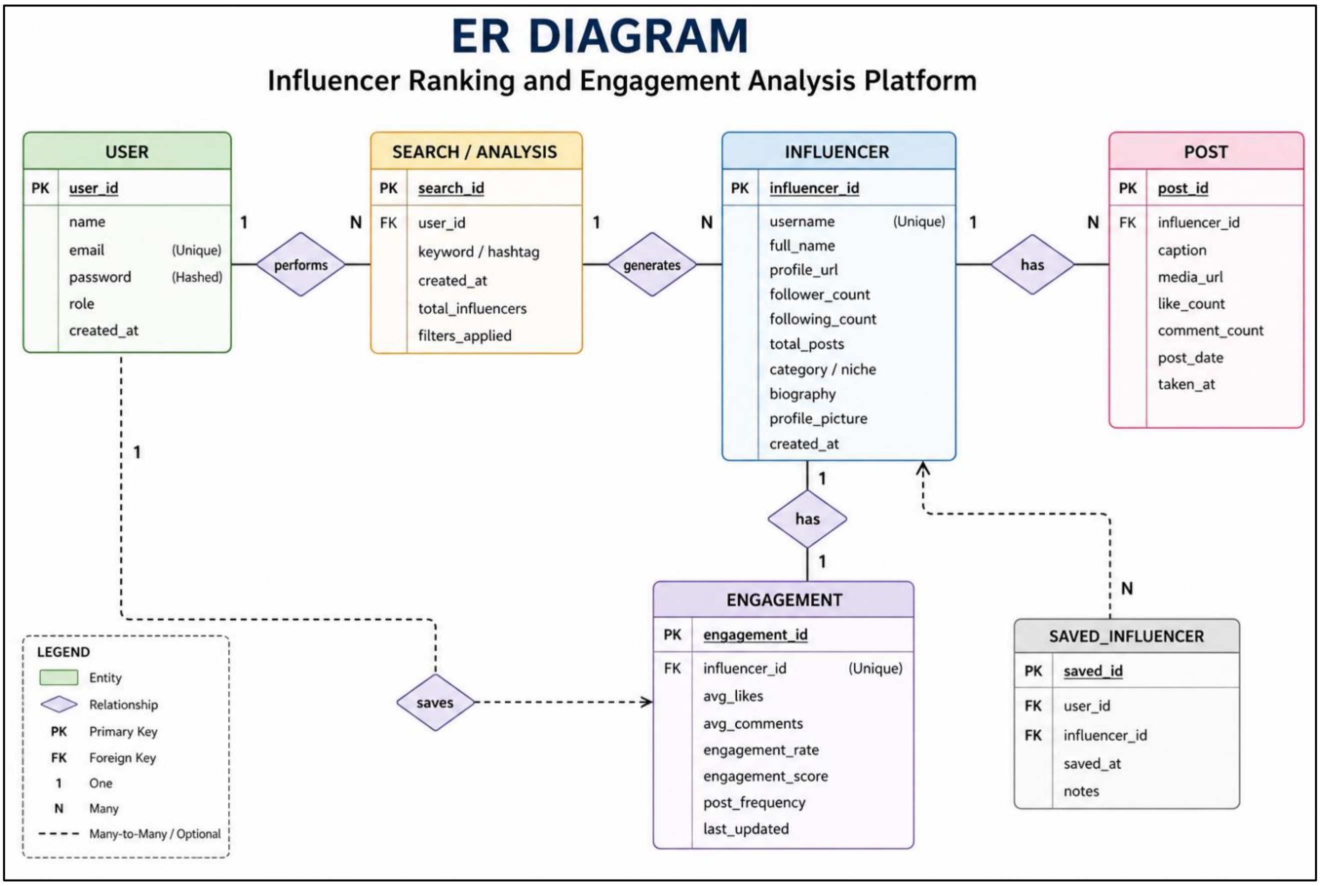
These relationships ensure proper linkage and data consistency within the system.

5.3 ER Diagram

The Entity Relationship Diagram (ERD) represents the database structure of the Influencer Ranking and Engagement Analysis Platform. It shows the entities such as User, Influencer, Post, and Engagement, along with their attributes and relationships.

The ER diagram helps in organizing data efficiently, maintaining relationships between different entities, and ensuring data consistency. It also provides a clear understanding of how data is stored and connected within the system.

ER Diagram :



CHAPTER-6

TECH STACK USED

6.1 Frontend: React.js

The frontend of the **Influencer Ranking and Engagement Analysis Platform** is developed using React.js, a popular JavaScript library for building dynamic and responsive user interfaces. React enables the creation of reusable components, which improves development efficiency and maintainability.

The frontend is responsible for:

- Accepting user inputs such as hashtags
- Displaying ranked influencers in a dashboard
- Providing filtering options for analysis
- Visualizing engagement metrics

Advantages of React.js:

- Component-based architecture for reusable UI
- Fast rendering using Virtual DOM
- Improved user experience with dynamic updates
- Easy integration with backend APIs

6.2 Backend: Node.js and Express.js

The backend is implemented using Node.js and Express.js, which provide a robust environment for handling server-side logic and API development.

The backend performs key operations such as:

- Handling requests from the frontend
- Integrating with the Apify API to fetch Instagram data

- Processing and analyzing engagement metrics
- Implementing the Engagement Score algorithm
- Managing user authentication and security

Advantages:

- Efficient handling of multiple requests
- Scalable and lightweight architecture
- Simplified API development using Express.js
- Easy integration with databases and external APIs

6.3 Database: MongoDB

MongoDB is used as the database to store influencer data, user information, and engagement metrics. It is a NoSQL database that stores data in a flexible JSON-like format.

Data stored includes:

- User details
- Influencer profiles
- Post data
- Engagement metrics and scores

Advantages:

- Flexible schema design
- High scalability for large datasets
- Fast data retrieval
- Seamless integration with Node.js

6.4 Tools and Deployment

Various tools are used during development and deployment to improve efficiency and maintain code quality.

- **VS Code:** Used for coding and debugging

- **Postman:** Used for testing APIs
- **GitHub:** Used for version control and code management
- **Deployment Platforms (e.g., Render/Vercel):** Used to host the application

These tools help in building, testing, and deploying the project effectively.

CHAPTER-7

RESULT

7.1 Influencer Ranking

The system successfully ranks influencers based on the calculated Engagement Score instead of relying on follower count. This approach highlights influencers with genuine audience interaction and helps identify top-performing creators for marketing campaigns.

- Ranked influencers based on engagement metrics
- Identified high-performing influencers in different categories
- Improved accuracy in influencer selection

7.2 Advanced Analysis Features

The platform provides advanced filtering and analysis capabilities to refine results based on user requirements.

- Filters based on follower count, engagement rate, and post count
- Removal of inactive or low-quality accounts
- Categorization of influencers based on engagement levels

7.3 Dashboard & Insights

An interactive dashboard is developed to present the results in a clear and user-friendly manner.

- Displays top influencers and engagement statistics
- Shows highest likes, comments, and follower insights
- Provides visual representation for better understanding

7.4 Data Management

The system efficiently manages analyzed data for future use.

- Saves analysis history
- Allows bookmarking of influencers

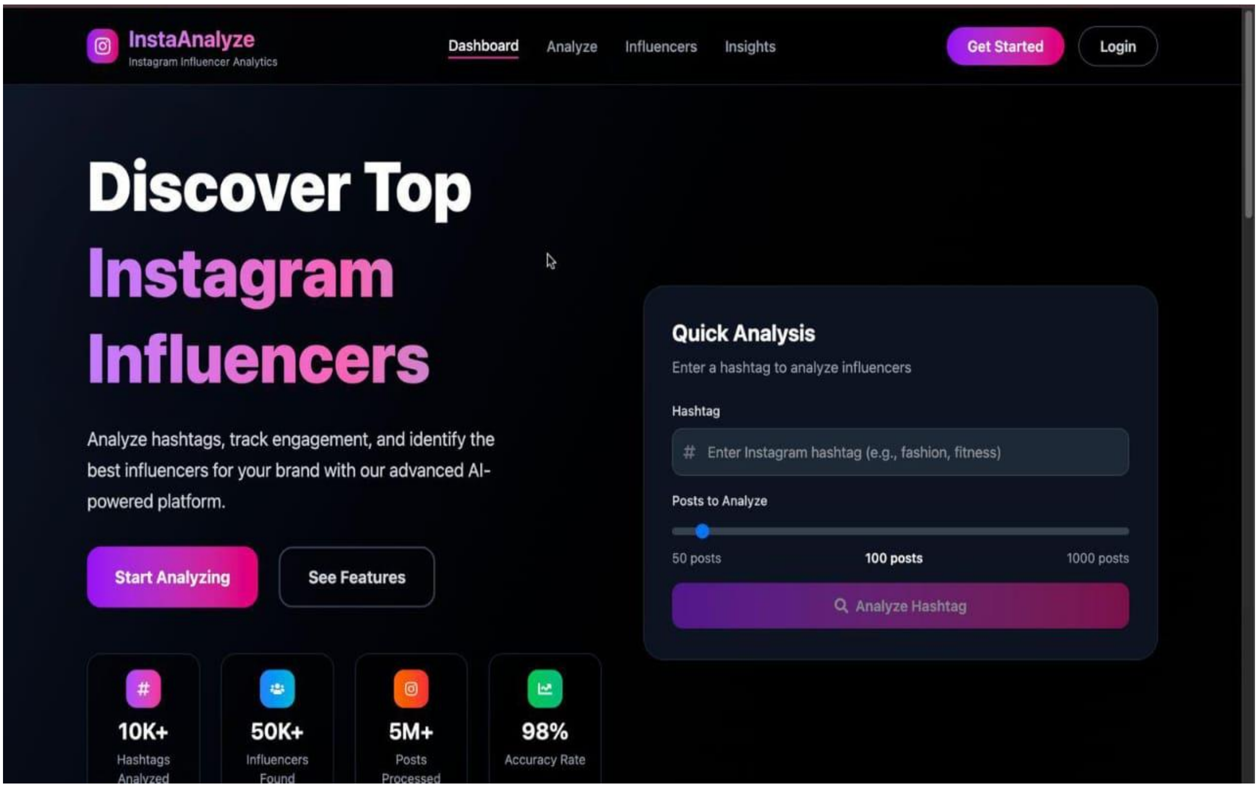
- Enables easy retrieval of previous results

7.5 Export Functionality

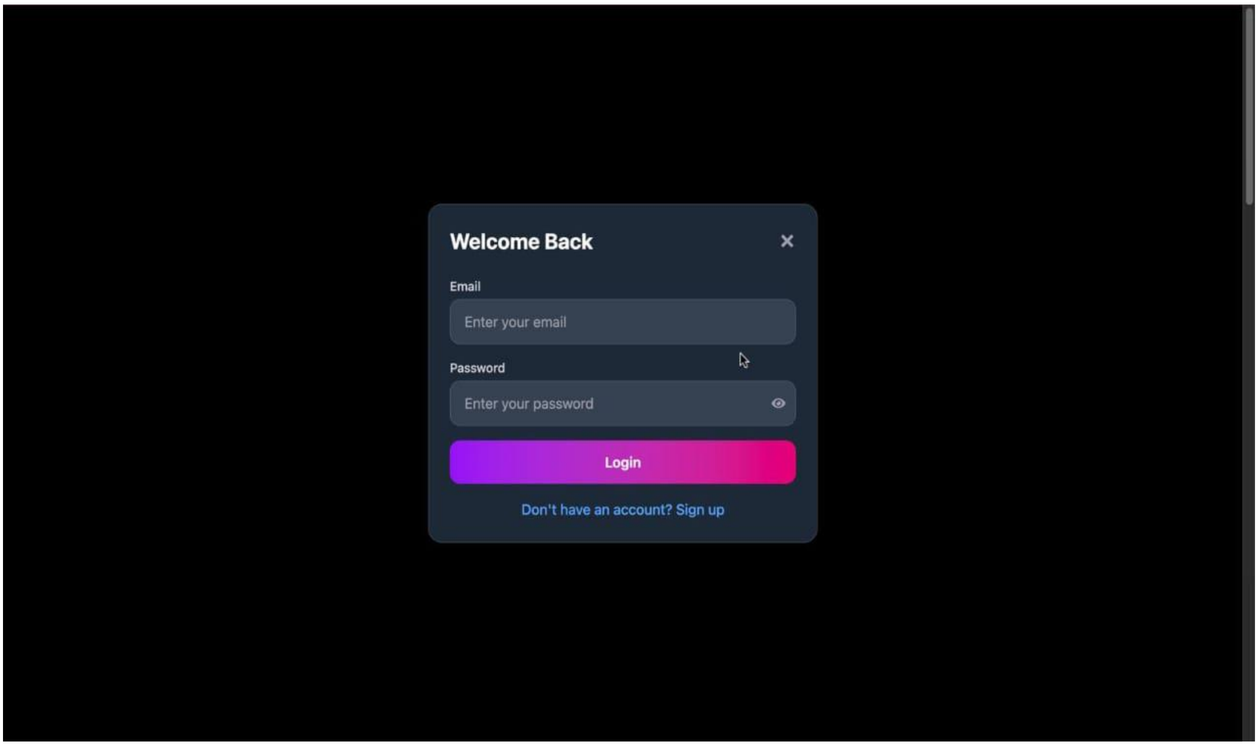
The platform supports exporting analyzed data for reporting and further processing.

- Export results in a structured format
- Useful for marketing teams and research purposes
- Helps in documentation and presentation

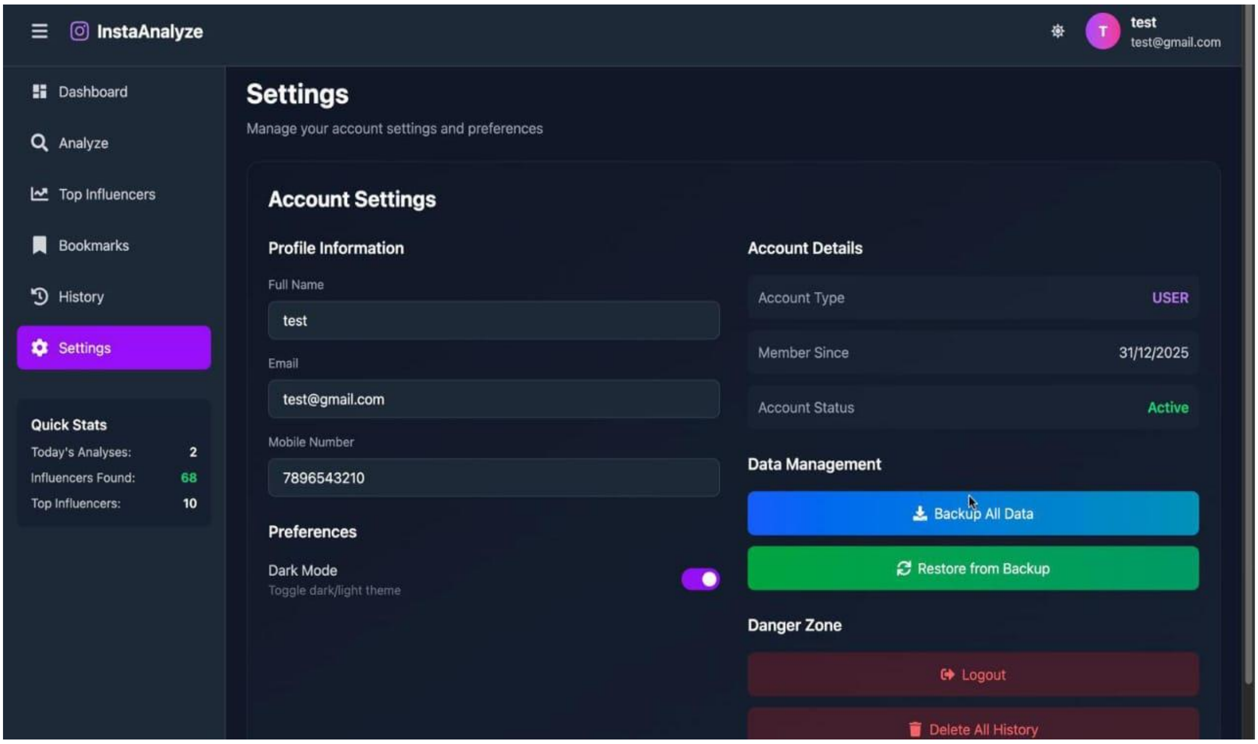
7.6 Screenshots of the Project



HOME PAGE

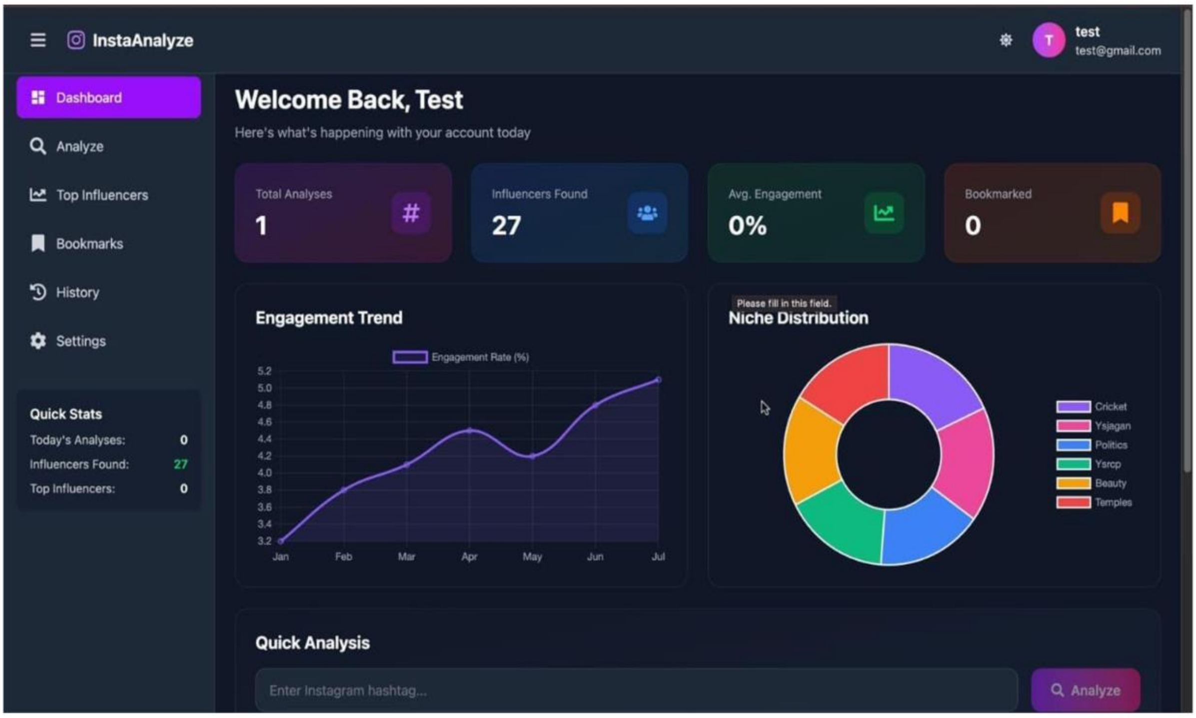


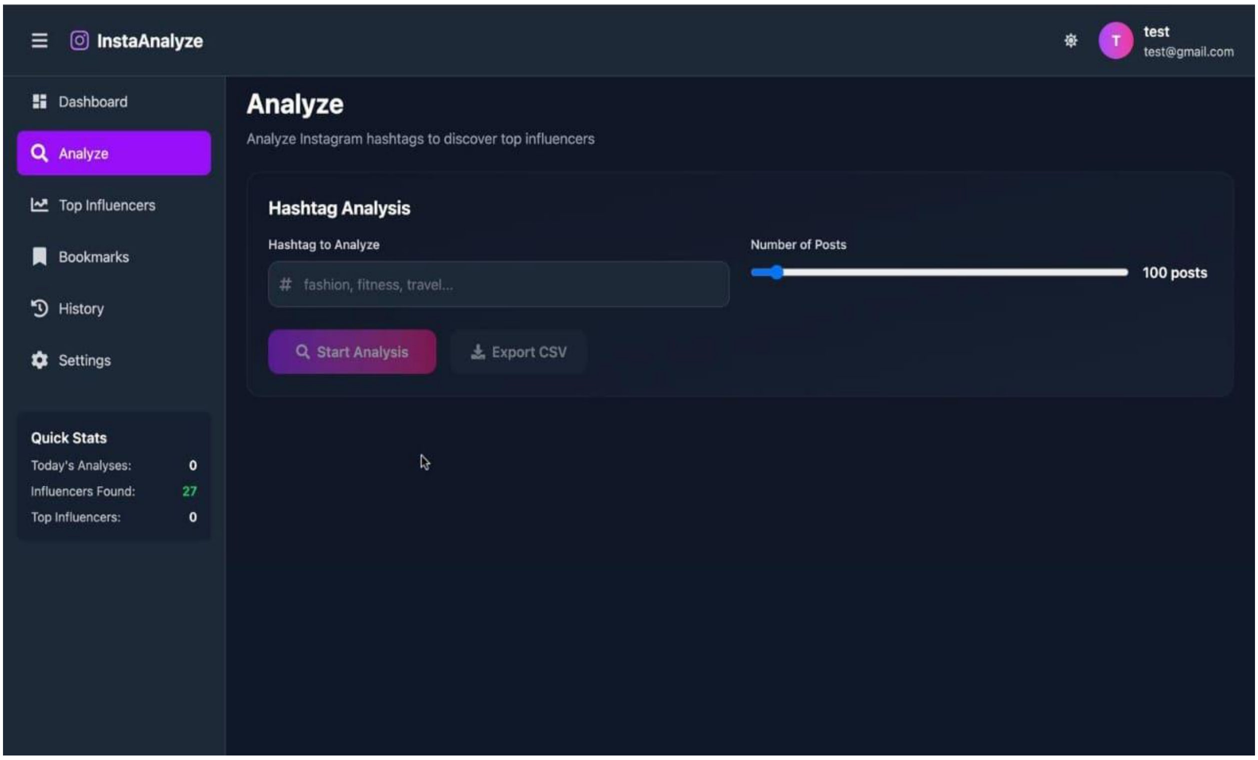
LOGIN PAGE



ACCOUNT SETTINGS

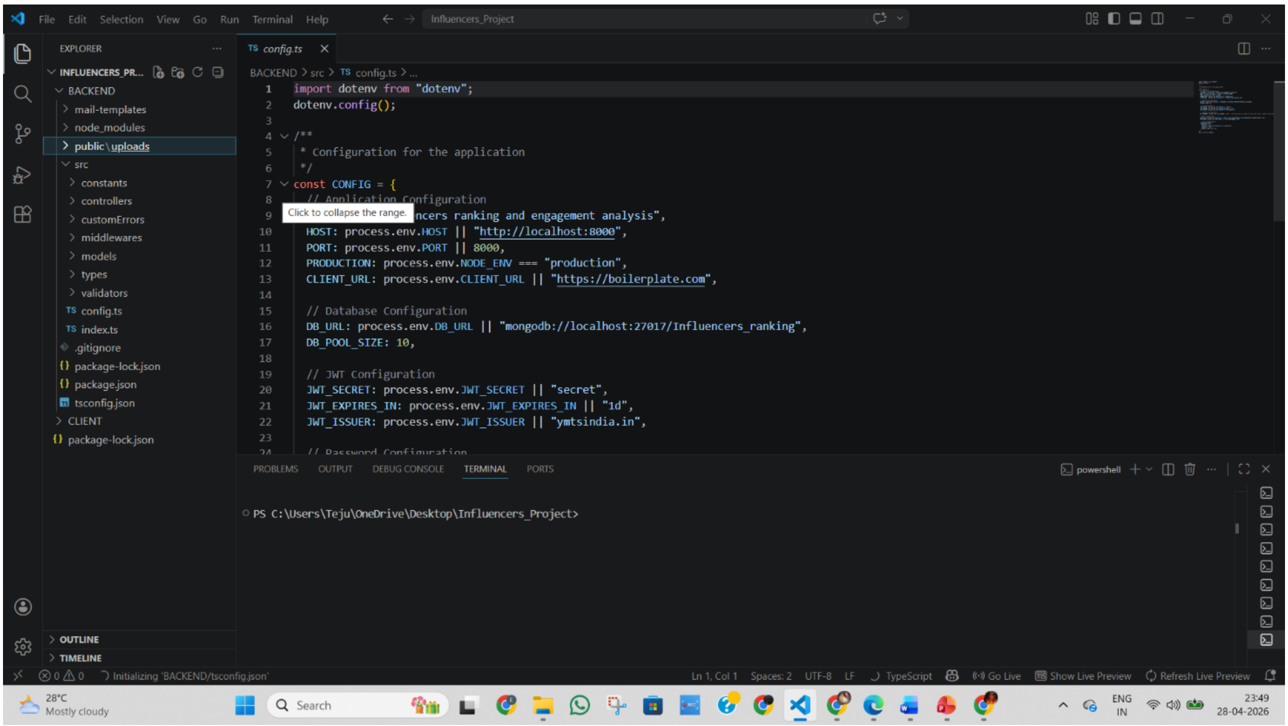
USER DASHBOARD

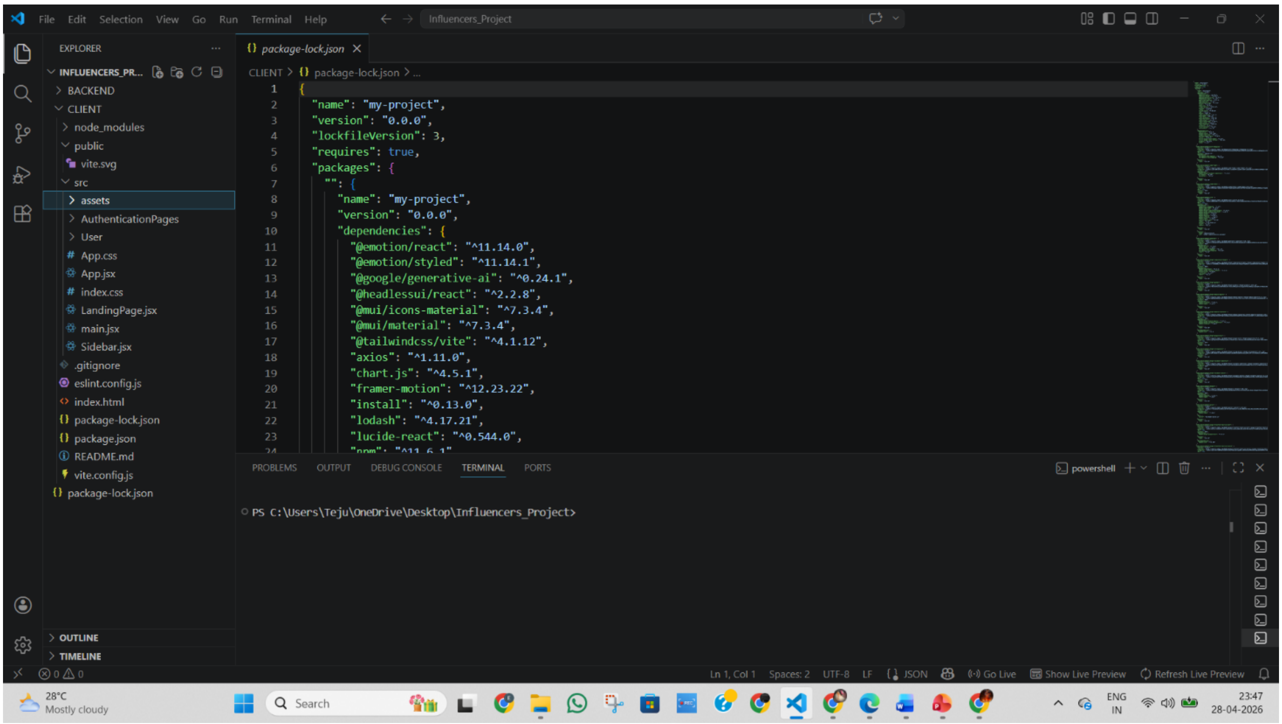


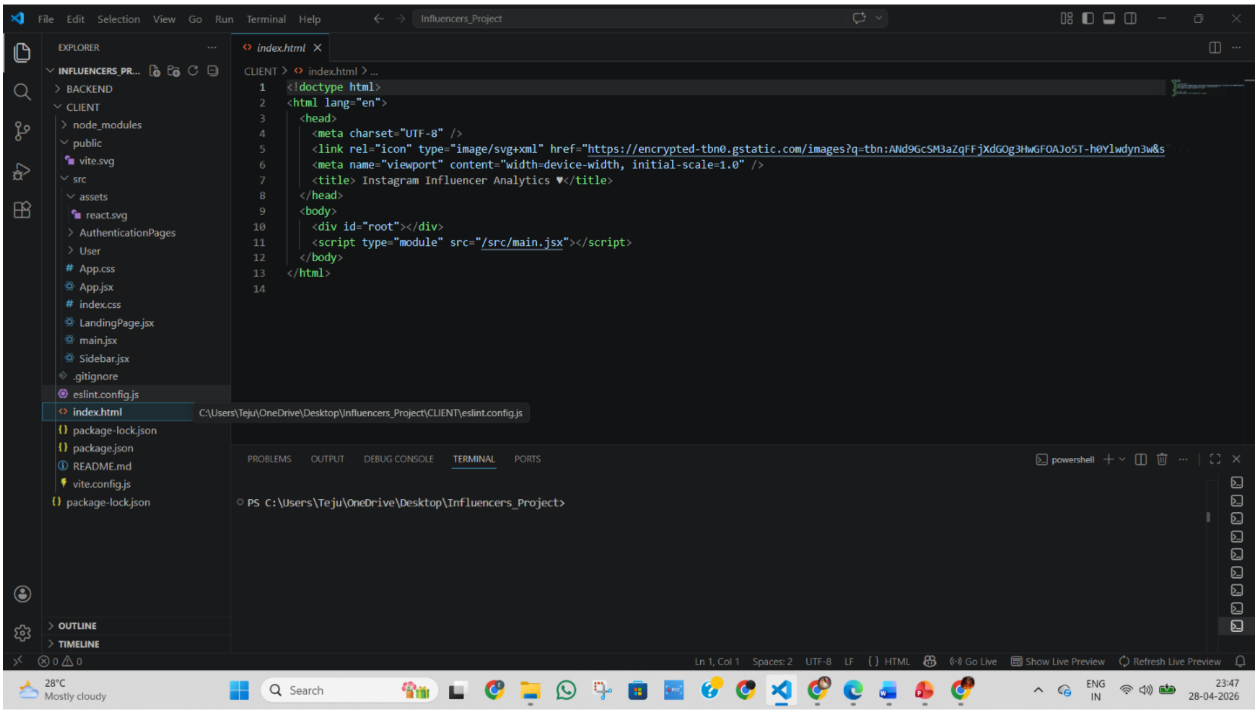


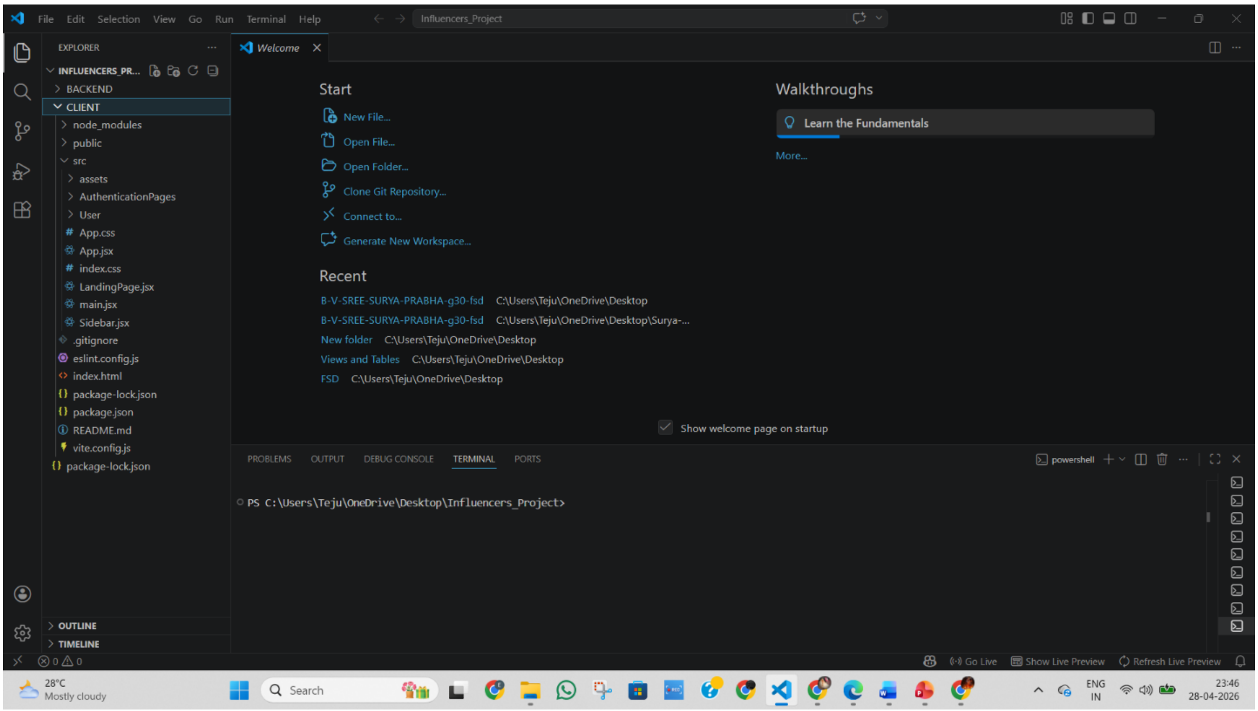
HASHTAG INPUT

Code Screenshots:









Conclusion:

The Influencer Ranking and Engagement Analysis Platform successfully provides an automated solution for identifying and ranking influencers based on real engagement metrics. The system eliminates dependency on follower count and introduces a data-driven approach to influencer evaluation.

By integrating real-time data collection, engagement analysis, and visualization features, the platform improves accuracy in decision-making and enhances marketing campaign effectiveness. It reduces manual effort, saves time, and provides reliable insights for selecting the most suitable influencers.

The project also demonstrates practical implementation of full-stack development using the MERN stack, making it valuable for both academic and real-world applications.

FUTURE ENHANCEMENTS

The system can be further improved by adding advanced features such as:

- Integration with multiple platforms (Instagram, YouTube, TikTok)
- AI/ML-based prediction of campaign performance
- Sentiment analysis of comments
- Detection of fake followers and bot activity
- Advanced analytics and reporting dashboards
- Complete influencer campaign management system